

project proposal

1. Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# read .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
```

```

aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# read in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# read in water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

```

2. Cleaning

sites object

```

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

Cleaning taxon_list

```

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

```

Cleaning water_quality

```

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names

```

```

clean_names() |>
# filtering join: only include sites in ncos_sites
semi_join(ncos_sites, by = "site") |>
# create new column to join with later data frames
unite("date_site", date, site, remove = TRUE) |>
# group by date_site column
group_by(date_site) |>
# calculate median pH, salinity, DO
summarize(
  med_ph = median(dissolved_oxygen_mg_l, na.rm = TRUE),
  med_sal = median(salinity_ppt, na.rm = TRUE),
  med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

```

Cleaning aq_ins

```

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(
  date, sample_type, site,
  ostracod,
  copepod,
  hemiptera_corixidae_boatman,
  cladocera,
  nematode,
  diptera_ceratopogonidae,
  annelida_oligochaete,
  diptera_chironomid,
  annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format

```

```

pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_do, .fun = "median", .na_rm = TRUE),
       site_full = fct_rev(site_full))

```

Figure 1: Invertebrate abundance as a function of DO

```

# create object sal_taxa from aq_ins_clean
DO_taxa <- aq_ins_clean |>
# remove rows that are missing median salinity values
filter(!is.na(med_do)) |>
# create column that calculates log-transformed abundance
mutate(log_ave_lit = log(ave_lit + 1)) |>
# mutates to title-case taxon names

```

```
mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
# order taxon levels by maximum average abundance per liter
mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE)) |>
# keep only the columns needed for the figure
select(date, site, site_full, taxon, ave_lit, log_ave_lit, med_do)

# display 10 random rows from sal_taxa
DO_taxa |> slice_sample(n = 10)
```

A tibble: 10 x 7

	date	site	site_full	taxon	ave_lit	log_ave_lit	med_do
	<dtm>	<chr>	<fct>	<fct>	<dbl>	<dbl>	<dbl>
1	2023-01-28 00:00:00	NPB1	North of Phelps B~	Clad~	0.267	0.236	9.74
2	2022-05-05 00:00:00	NEC	South of Venoco B~	Dipt~	0	0	7.3
3	2023-03-03 00:00:00	NPB2	Phelps Road	Dipt~	0.267	0.236	9.59
4	2023-06-06 00:00:00	NPB1	North of Phelps B~	Nema~	0	0	2.38
5	2020-02-21 00:00:00	NVBR	North of Venoco B~	Nema~	0	0	6.1
6	2023-06-06 00:00:00	NPB1	North of Phelps B~	Anne~	0.267	0.236	2.38
7	2023-06-06 00:00:00	NPB1	North of Phelps B~	Dipt~	2.67	1.30	2.38
8	2023-03-03 00:00:00	NDC	Devereux Creek	Nema~	0	0	5.64
9	2023-09-15 00:00:00	NMC	Main Channel	Anne~	0	0	14.2
10	2023-05-16 00:00:00	NVBR	North of Venoco B~	Anne~	0	0	9.83

```
# base layer: ggplot
DO_taxa_plot <- ggplot(data = DO_taxa, # assigns plot to object do_taxa_plot
# uses sal_taxa data frame
mapping = aes(x = med_do, # x-axis: med_do
y = log_ave_lit, # y-axis: log_ave_lit
color = taxon)) + # color by taxon

# first layer: points show individual abundance measurements
geom_point(shape = 16, # shape is circle
size = 2, # size is 2
alpha = 0.7) + # opacity is 0.7

# separate panels for each taxon with free x- and y-axis scales
facet_wrap(~ taxon, scales = "free") +

# use non-default colors for points
scale_color_viridis_d() +

labs(x = "Median DO (mg/L)", # relabel x-axis
y = "log + 1 transformed average abundance/L", # relabel y-axis
# creates descriptive title
title = "Invertebrate abundance as a function of DO") +
```

```
# uses non-default theme
theme_minimal() +
# remove redundant legend
theme(legend.position = "none")

# shows plot
DO_taxa_plot
```

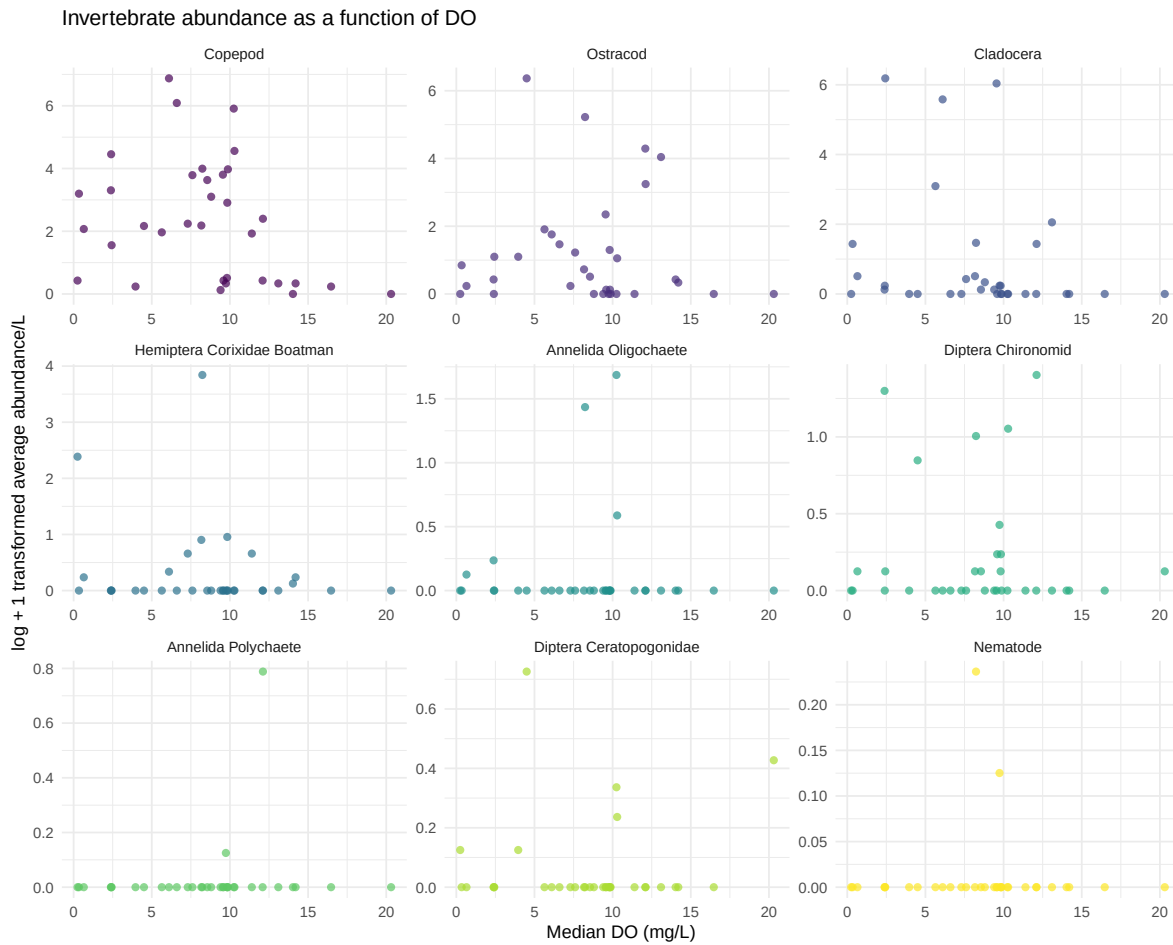


Figure 2: Variance of DO across sights

```
# base layer
DO <- ggplot(data = aq_ins_clean,
```

```

# x-axis: site
# y-axis: mean salinity
mapping = aes(x = site_full,
              y = med_do)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
            color = "red",
            size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
     y = "Median DO (mh/L)",
     title = "Site DO levels") +
# different theme
theme_bw()
# show plot
DO

```

